Stress-Testing Product-Gated Clustering and Bounded Priority Ranking for Flood-Rescue Reports: A Synthetic Study

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ISDS 2026 | Paper 6444 |

Presentation overview

From noisy reports to measured scheduling consequences

01 **Problem and research questions**
Why consolidation errors matter

02 **Methods and observable pipeline**
Graph, clustering and bounded priority

03 **Experimental design and metrics**
Two separate suites; run-level evaluation

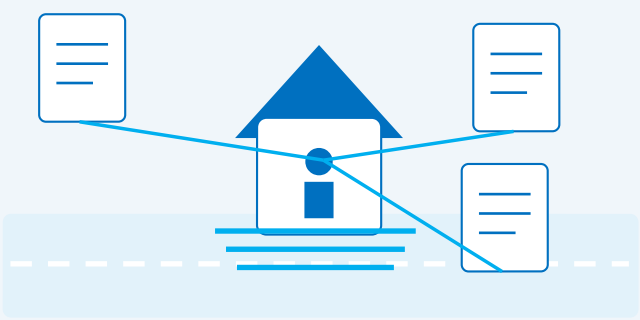
04 **Results for RQ1–RQ3**
Clustering → ranking → dispatch

05 **Contributions and conclusions**
What is proved, observed and still open

Many reports do not mean many incidents

Schematic flood scenarios: repeated reports, nearby incidents and false destinations

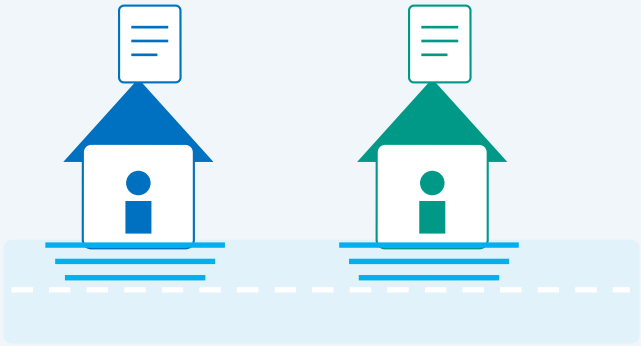
Repeated reports



Many reports,
one physical incident

Risk: repeated trips

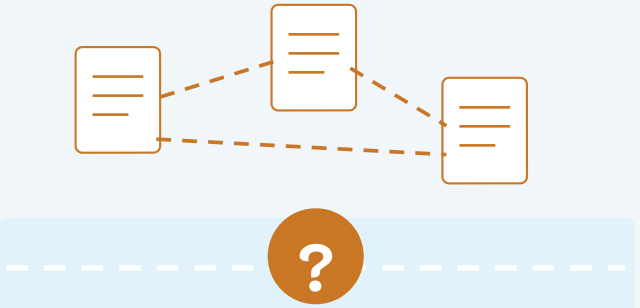
Nearby incidents



Close in space,
different incidents

Risk: a missed incident

False reports



Several reports,
no genuine incident

Risk: a false destination

Related approaches and the study gap

Three research strands meet at the report-to-dispatch boundary

Social sensing

Event detection and crisis-information extraction

[6–8]

Report consolidation

Graph community detection and density-based clustering

[1]; Paper 6444 §1

Humanitarian dispatch

Competing need, efficiency and equity objectives

[5]

GAP EXAMINED IN THIS STUDY

How do predicted splits, merges and false destinations propagate through ranking into a scheduler that cannot consult incident truth?

Research questions: do safeguards improve dispatch?

Three stage-specific questions guide the methods, data and evaluation

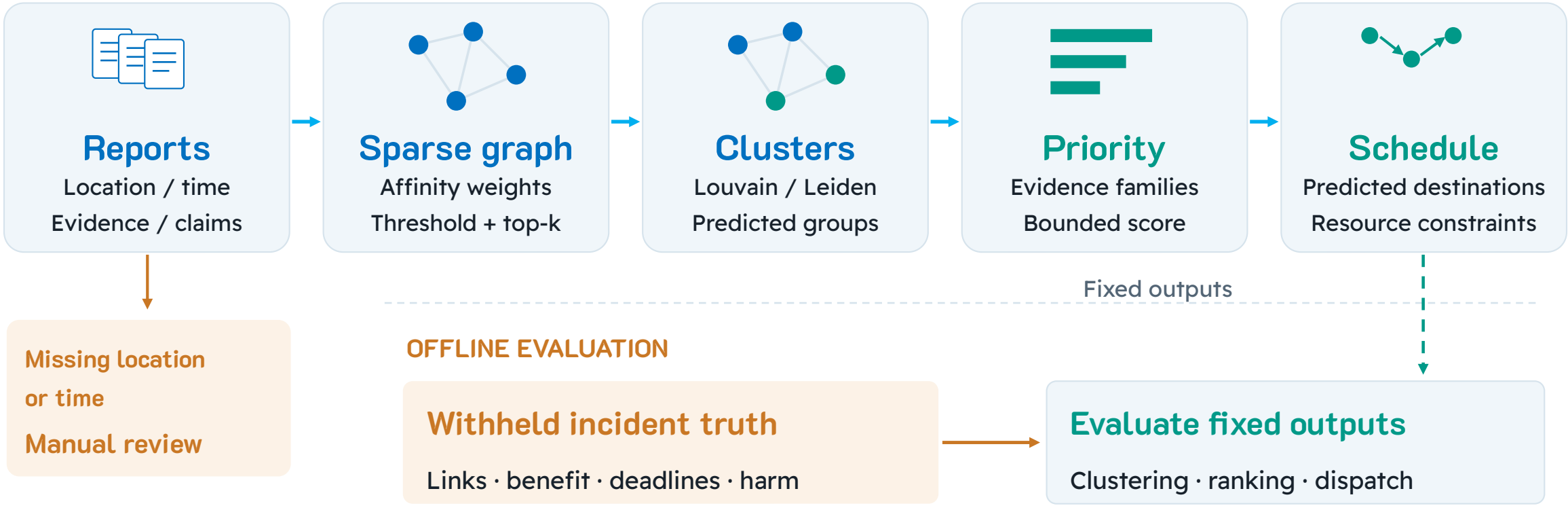
A safeguard at one stage does not establish benefit at the next.

RQ1	Clustering	Does the tuned Product pipeline improve clustering over Additive?
RQ2	Ranking	Does duplicate-aware priority align with independent simulated benefit?
RQ3	Dispatch	How do predicted splits, merges and noise affect dispatch outcomes?

Observable pipeline: reports to a fixed schedule

Algorithm inputs and evaluator-only truth stay separate

ALGORITHM · OBSERVABLE INPUTS ONLY



Generator 3.0: what does one run contain?

RQ1 · geographically anchored, semi-synthetic · 80 runs · 26,288 reports

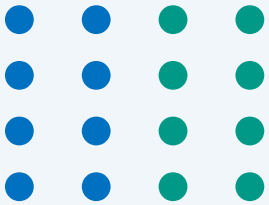
Copernicus EMSR848
Historical flood context

OpenStreetMap
Geographic anchors

WorldPop 2025
Population reference

1 RUN = ONE GENERATED SCENARIO

16 latent incidents



Simulated truth

288–370 reports



Genuine + copies + campaigns

Two separate exports

Inference: observable reports

Evaluation: withheld labels

Why two generators? Different questions need different truth

Candidate 4.1 is fully synthetic and independent of the Generator-3.0 sample

Generator 3.0 80 runs · 16 incidents/run 26,288 reports	→ **Required evaluator truth** Incident linkage + controlled graph stresses	→ **RQ1** Clustering
Candidate 4.1 80 datasets · 16 incidents each 30,229 reports	→ **Required evaluator truth** Independent benefit, deadlines, service + harm	→ **RQ2 / RQ3** Ranking / dispatch

Separate samples: complementary stage tests, not one common-sample validation chain.

Why test Product? Geography gates the whole affinity

A plausible mechanism to test; a geographic edge bound alone does not ensure better clusters

PRODUCT

$$w_{ij}^{\times} = a_i a_j G_{ij}(\beta T_{ij} + \gamma C_{ij})$$

ADDITIVE

$$w_{ij}^{+} = a_i a_j(\alpha G_{ij} + \beta T_{ij} + \gamma C_{ij})$$

G Geographic proximity
T Temporal proximity
C Context similarity
a Location/time admission

alpha, beta, gamma: nonnegative weights

ILLUSTRATIVE PAIR · SAME TIME AND CONTEXT, FAR APART



Report A Report B

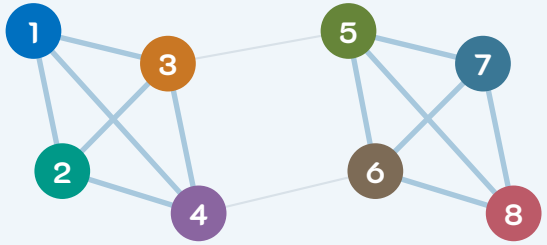
G = .02; T = C = .90; a = 1; all weights = 1

Product .036 **Additive 1.820**

Louvain: local moves and graph aggregation

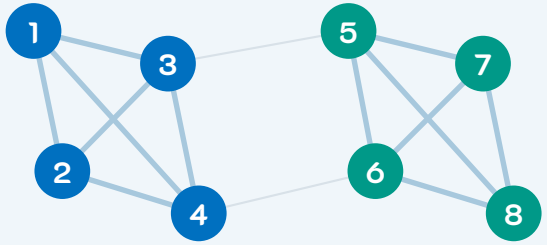
Schematic weighted network · modularity is an objective, not incident-label accuracy

1 Weighted graph



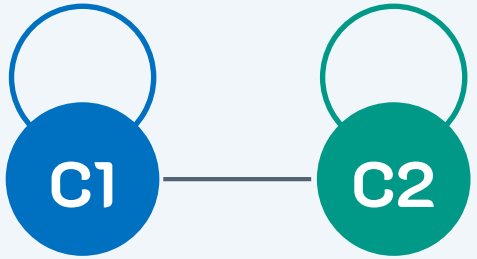
Start with one community for each report node.

2 Local moves



Move nodes when modularity increases.

3 Aggregate



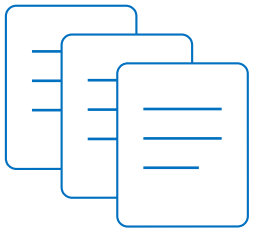
Between-group weight

Communities become weighted super-nodes.

Repeat on the smaller graph until no further modularity gain is obtained.

Bounded priority limits copies, not dishonest claims

Duplicate handling and information verification are different tasks



Repeated payload

Evidence families

Exact / near-copy grouping
Family maxima

Bounded score

Confidence, clipping
and saturation

$$0 \le P_k \le \mu, \qquad 1 \le \mu \le 2$$

Exact-copy invariance

Requires fixed families and score inputs.

Inflated headcounts remain unverified

Clipping caps influence; it does not verify a claim.

Evaluation protocol and information boundaries

Select on calibration; hold configurations fixed for the test audit

Generator 3.0 / RQ1

20 development	20 calibration	40 held-out test

Candidate 4.1 / RQ2–RQ3

20 development	20 calibration	40 held-out test

RQ2: oracle incident grouping

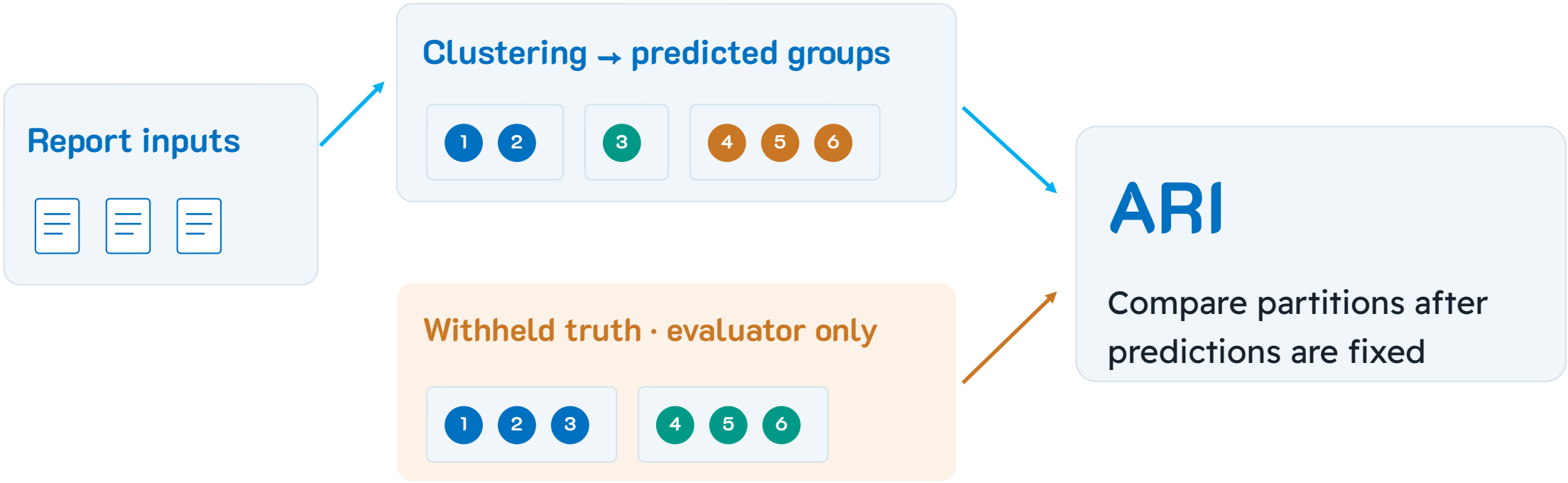
Isolate ranking from upstream clustering errors.

RQ3: predicted clusters

Average resource conditions within each seed: 40 pairs.

One held-out run: compare groups with withheld truth

ARI measures partition agreement, corrected for chance · schematic partition example



ARI: 1 = identical; ≈ 0 = chance; negative values are possible. Higher is better.

Pairwise F1 complements ARI by balancing false merges and false splits.

Reported metrics: ranking and score stability

Normalized Discounted Cumulative Gain at five (NDCG@5) and score stability

NDCG@5 ↑

$$\mathrm{DCG@5} = \sum_{r=1}^{5} \frac{b_{\pi_r}}{\log_2(r+1)}$$

$$\mathrm{NDCG@5} = \frac{\mathrm{DCG@5}}{\text{Ideal DCG@5}}$$

b: independent simulated benefit

π: order induced by the priority score

1: ideal ranking; larger is better

Priority drift ↓

Absolute normalized score change after a perturbation.

Top-five churn ↓

Change in membership of the five highest-ranked incidents.

Reported metrics: dispatch outcomes

Consequences are evaluated only after the schedule is fixed

Simulated harm

LOWER IS BETTER

Mean incident harm from the synthetic lateness / service model, not measured real-world injury.

Deadline misses

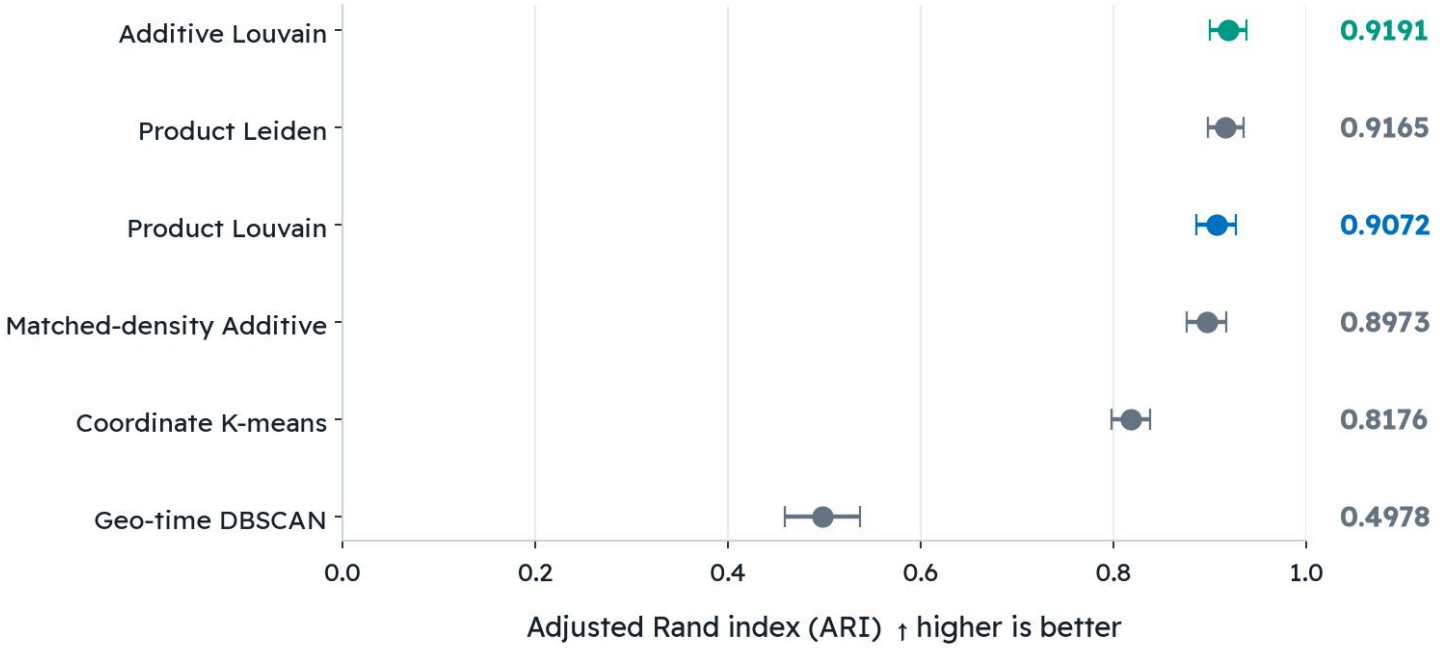
LOWER IS BETTER

Fraction of incidents whose simulated response time exceeds their own deadline.

Travel / service constraints → Locked schedule → Evaluator-only outcomes

RQ1: no Product composition advantage is established

Held-out Generator-3.0 test split · 40 paired runs



Additive Louvain

0.9191

Product Louvain

0.9072

Paired Holm p = .2560

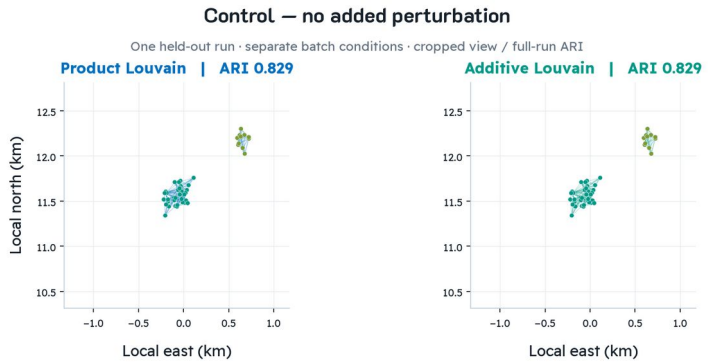
16.49%

Product false destinations

RQ1: fivefold copies expose a major failure case

Static 40-run result plus one-run illustration · separate fixed batch stresses

ONE-RUN ILLUSTRATION



5× copies · Product ΔARI

−0.6108

95% CI [−.6318, −.5881]

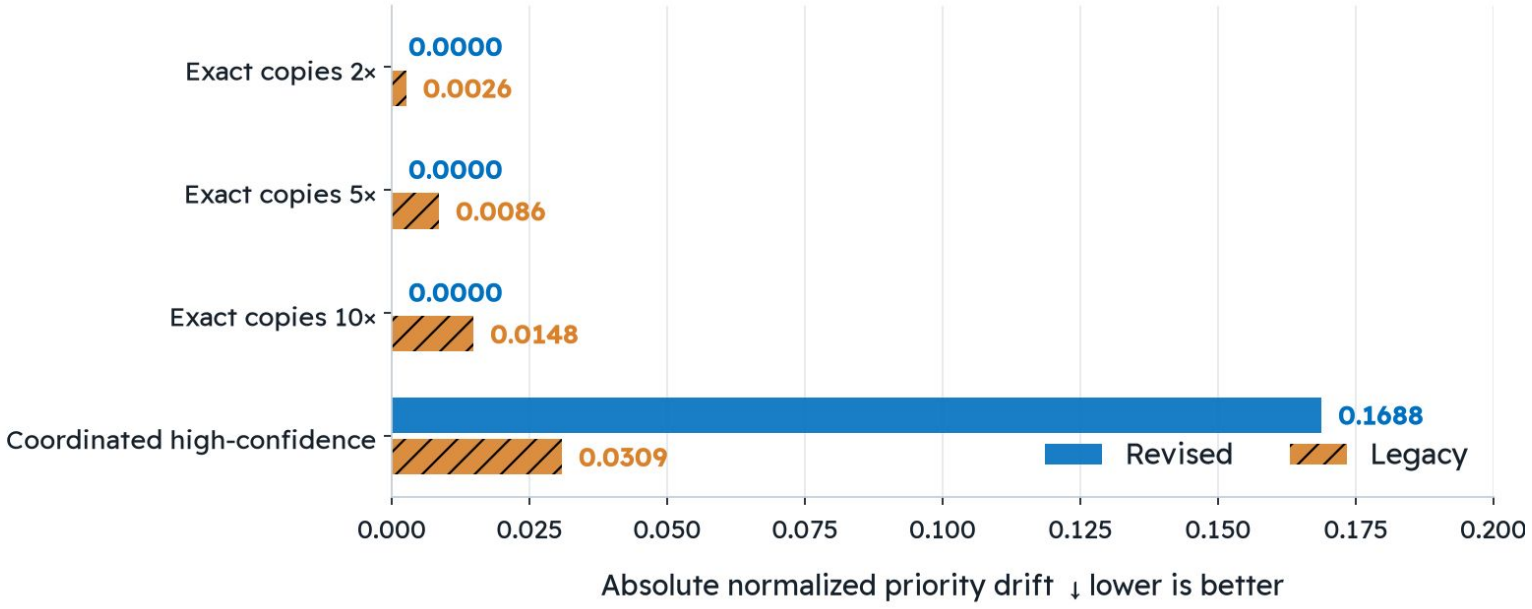
Mean paired change in ARI

40 held-out runs · negative = degradation

	GPS 100 m	GPS 300 m	Time 15 min	Time 60 min	Copies 2×	Copies 5×
Product Louvain	-0.012	-0.126	-0.035	-0.163	+0.024	-0.611
Additive Louvain	-0.022	-0.156	-0.038	-0.154	+0.015	-0.622
MD Additive	-0.019	-0.098	-0.044	-0.190	+0.021	-0.606
Product Leiden	-0.012	-0.125	-0.031	-0.156	+0.015	-0.620
Geo-time DBSCAN	-0.030	-0.127	-0.012	-0.055	+0.000	+0.000

RQ2: exact-copy stability has a narrow scope

Oracle incident groups and fixed score inputs



Exact copies

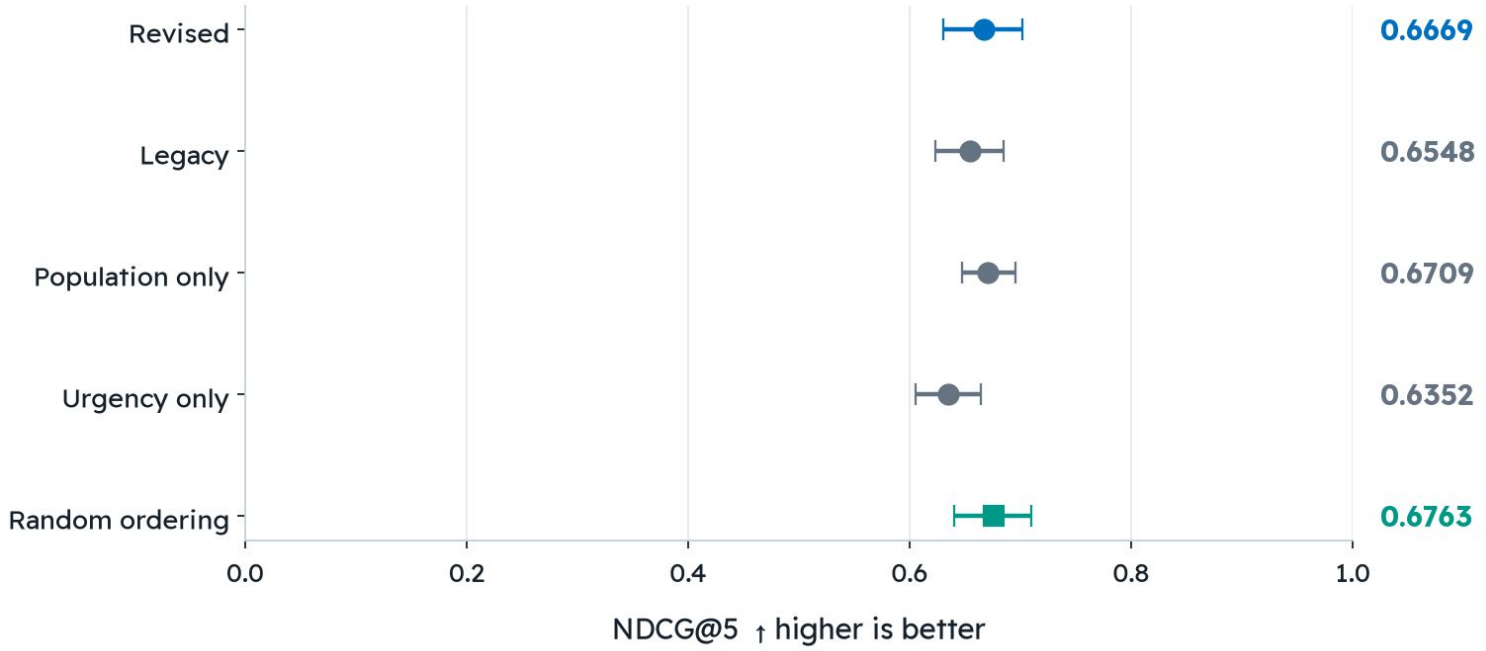
2× / 5× / 10×:
zero revised score drift.

Coordinated attack

High-confidence payloads
still distort priority.

RQ2: alignment with simulated benefit remains weak

Even oracle grouping does not establish a general ranking advantage



Revised NDCG@5

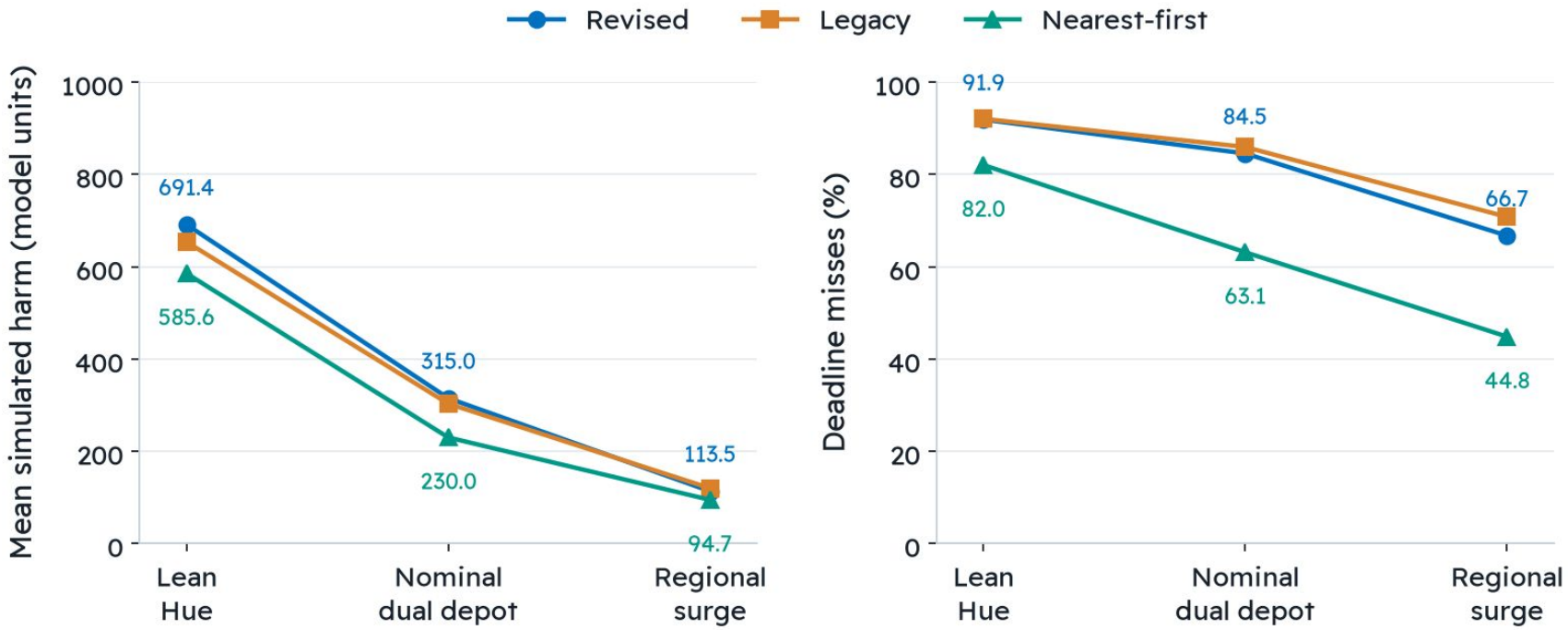
0.6669

Random comparator

0.6763

RQ3: safeguards do not ensure better dispatch

Product partition · three resource conditions · 40 test seeds



Three contributions, with explicit evidence limits

Mathematical guarantees, simulated findings and unresolved validation are different claims

CONTRIBUTION	MATHEMATICALLY PROVED	OBSERVED IN SIMULATION	NOT YET ESTABLISHED
1 Product graph	Conditional retained-edge localization	No composition advantage established	Cluster compactness or transfer advantage
2 Family-wise priority	Copy invariance with fixed families / inputs	Zero copy drift; weak benefit alignment	Claim authenticity or true headcounts
3 Downstream audit	No dispatch-policy guarantee	Paired audit exposes adverse outcomes	Expert priorities or field effectiveness

Conclusion: stage safeguards do not establish dispatch benefit

Conditional theory and a synthetic audit identify what still needs validation

A valid local guarantee is only one part of the report-to-dispatch problem.

Clustering

Conditional edge bound; no Product advantage established.

→

Ranking

Exact-copy stability; benefit alignment remains weak.

→

Dispatch

Stage safeguards do not ensure improved simulated outcomes.

Future work: close the evidence gaps

Proposed validation steps, not results already obtained

01 Real report linkage

Acquire traceable report snapshots and independently annotate same-incident links.

02 Expert and source validation

Verify headcounts and source credibility; assess priorities with rescue experts.

03 Transfer and stress coverage

Freeze the pipeline and test new event contexts, missingness and source-quality mechanisms.

04 Scheduling objectives

Evaluate need-based priorities together with travel, service capacity and resource constraints.

References (1 of 2)

Methods and evaluation

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Data and application context

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THANK YOU

Questions and discussion

Clustering safeguards, ranking stability and dispatch benefit require separate evaluation.

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Research artifact [12]: github.com/ngthtrong/nckh2/releases/tag/v1.0.1

Funded by Can Tho University · THS2026-68